

Representative Structural Calculation Package

Open parking frame

Item	Value
Project	Open parking frame
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	06-open-parking-frame

Nodes	Members	Loads	Combos	Max disp.	Max util.
72	145	410	28	13.55 mm	1.045

Model And Design Basis

Item	Value
Building type	parking
Description	Five-story open parking structure with repetitive long bays.
Occupancy	Parking
Bounds X/Y/Z	22.50 / 15.00 / 16.00
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	85	0.000 kN	0.000 kN	-5400 kN
L	live	85	0.000 kN	0.000 kN	-2734 kN
WX	wind	60	108.00 kN	0.000 kN	0.000 kN
WY	wind	60	0.000 kN	162.00 kN	0.000 kN
EX	seismic	60	517.09 kN	0.000 kN	0.000 kN
EY	seismic	60	0.000 kN	517.09 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	337.50	1080 kN	607.50 kN	21.60 kN	32.40 kN
2	337.50	1080 kN	607.50 kN	21.60 kN	32.40 kN
3	337.50	1080 kN	607.50 kN	21.60 kN	32.40 kN
4	337.50	1080 kN	607.50 kN	21.60 kN	32.40 kN
5	337.50	1080 kN	303.75 kN	21.60 kN	32.40 kN

Load Derivation Formula Trace

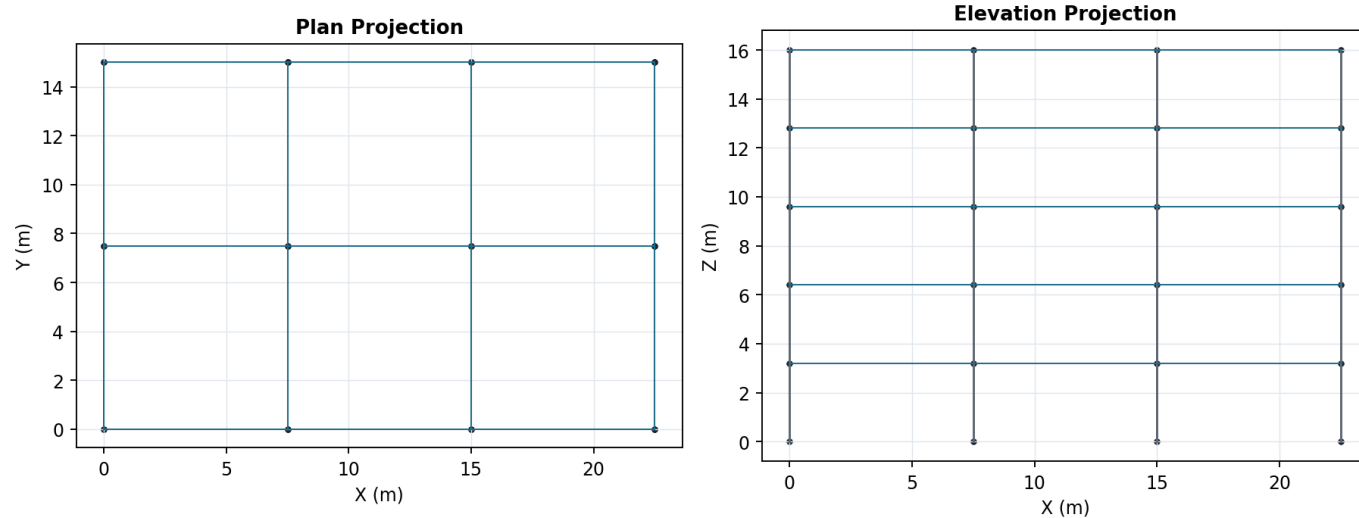
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=parking
basis-plan-area	-	$\max(\text{model width} * \text{model depth}, 1)$	337.50 m2	Bx=22.50 m, By=15.00 m
basis-roof-area	-	roofArea input or typical floor area	337.50 m2	roofArea=337.50 m2
D-ST1	D	$A * q_D$	1080 kN	$A=337.50 \text{ m}^2$, $q_D=3.200 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	607.50 kN	$A=337.50 \text{ m}^2$, $q_L=1.800 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	21.60 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=15.00 \text{ m}$, $h=3.200 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	32.40 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=22.50 \text{ m}$, $h=3.200 \text{ m}$
D-ST2	D	$A * q_D$	1080 kN	$A=337.50 \text{ m}^2$, $q_D=3.200 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	607.50 kN	$A=337.50 \text{ m}^2$, $q_L=1.800 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	21.60 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=15.00 \text{ m}$, $h=3.200 \text{ m}$
WY-ST2	WY	$p_{WY} * B_x * h$	32.40 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=22.50 \text{ m}$, $h=3.200 \text{ m}$
D-ST3	D	$A * q_D$	1080 kN	$A=337.50 \text{ m}^2$, $q_D=3.200 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	607.50 kN	$A=337.50 \text{ m}^2$, $q_L=1.800 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	21.60 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=15.00 \text{ m}$, $h=3.200 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

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Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	2.540 mm	0.235	0.000
KDS-ST-02	OK	3.445 mm	0.337	0.000
KDS-ST-04-WX-P	OK	3.545 mm	0.330	0.000
KDS-ST-04-WX-N	OK	3.545 mm	0.330	0.000
KDS-ST-04-WY-P	OK	4.395 mm	0.356	0.000
KDS-ST-04-WY-N	OK	4.395 mm	0.356	0.000
KDS-ST-05-EX-P	OK	12.53 mm	0.505	0.000
KDS-ST-05-EX-N	OK	12.53 mm	0.505	0.000
KDS-ST-05-EY-P	OK	13.55 mm	0.518	0.000
KDS-ST-05-EY-N	OK	13.55 mm	0.518	0.000
KDS-ST-06-WX-P	OK	2.627 mm	0.195	0.000
KDS-ST-06-WX-N	OK	2.627 mm	0.195	0.000
KDS-ST-06-WY-P	OK	3.703 mm	0.221	0.000
KDS-ST-06-WY-N	OK	3.703 mm	0.221	0.000
KDS-ST-07-EX-P	OK	12.29 mm	0.371	0.000
KDS-ST-07-EX-N	OK	12.29 mm	0.371	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	3.604 mm
Max drift ratio	0.00113
Governing	KDS-ST-05-EY-N / story 2

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.200 m	0.018 mm	0.00001	0.001	OK
KDS-ST-01	2	3.200 m	0.019 mm	0.00001	0.001	OK
KDS-ST-01	3	3.200 m	0.000 mm	0.00000	0.000	OK
KDS-ST-01	4	3.200 m	0.020 mm	0.00001	0.001	OK
KDS-ST-01	5	3.200 m	0.081 mm	0.00003	0.005	OK
KDS-ST-02	1	3.200 m	0.026 mm	0.00001	0.002	OK
KDS-ST-02	2	3.200 m	0.028 mm	0.00001	0.002	OK
KDS-ST-02	3	3.200 m	0.002 mm	0.00000	0.000	OK
KDS-ST-02	4	3.200 m	0.015 mm	0.00000	0.001	OK
KDS-ST-02	5	3.200 m	0.095 mm	0.00003	0.006	OK
KDS-ST-04-WX-P	1	3.200 m	0.440 mm	0.00014	0.027	OK
KDS-ST-04-WX-P	2	3.200 m	0.616 mm	0.00019	0.038	OK
KDS-ST-04-WX-P	3	3.200 m	0.492 mm	0.00015	0.031	OK
KDS-ST-04-WX-P	4	3.200 m	0.351 mm	0.00011	0.022	OK
KDS-ST-04-WX-P	5	3.200 m	0.263 mm	0.00008	0.016	OK
KDS-ST-04-WX-N	1	3.200 m	0.440 mm	0.00014	0.027	OK
KDS-ST-04-WX-N	2	3.200 m	0.616 mm	0.00019	0.038	OK
KDS-ST-04-WX-N	3	3.200 m	0.492 mm	0.00015	0.031	OK
KDS-ST-04-WX-N	4	3.200 m	0.351 mm	0.00011	0.022	OK
KDS-ST-04-WX-N	5	3.200 m	0.263 mm	0.00008	0.016	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M55	NG	1.045	steel-deflection	KDS-ST-05-EY-N
2	M54	NG	1.045	steel-deflection	KDS-ST-05-EY-P
3	M59	NG	1.042	steel-deflection	KDS-ST-05-EX-N
4	M51	NG	1.042	steel-deflection	KDS-ST-05-EX-N
5	M58	NG	1.042	steel-deflection	KDS-ST-05-EX-P
6	M50	NG	1.042	steel-deflection	KDS-ST-05-EX-P
7	M56	NG	1.040	steel-deflection	KDS-ST-05-EY-N
8	M53	NG	1.040	steel-deflection	KDS-ST-05-EY-N
9	M60	NG	1.038	steel-deflection	KDS-ST-05-EY-P
10	M52	NG	1.038	steel-deflection	KDS-ST-05-EY-N

11	M49	NG	1.038	steel-deflection	KDS-ST-05-EY-N
12	M57	NG	1.038	steel-deflection	KDS-ST-05-EY-P
13	M43	WARN	0.926	steel-deflection	KDS-ST-05-EY-N
14	M42	WARN	0.926	steel-deflection	KDS-ST-05-EY-N
15	M39	WARN	0.922	steel-deflection	KDS-ST-05-EX-P
16	M47	WARN	0.922	steel-deflection	KDS-ST-05-EX-P
17	M38	WARN	0.922	steel-deflection	KDS-ST-05-EX-N
18	M46	WARN	0.922	steel-deflection	KDS-ST-05-EX-N
19	M44	WARN	0.920	steel-deflection	KDS-ST-05-EY-P
20	M41	WARN	0.920	steel-deflection	KDS-ST-05-EY-P

Member Design Trace Matrix

Item	Value
Checked members	145
Unimplemented members	0
NG / WARN	12/28
Max utilization	1.045

Member	Type	Status	Util.	Formula	Action
M55	steel	NG	1.045	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M54	steel	NG	1.045	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M59	steel	NG	1.042	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M51	steel	NG	1.042	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M58	steel	NG	1.042	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M50	steel	NG	1.042	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M56	steel	NG	1.040	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M53	steel	NG	1.040	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M60	steel	NG	1.038	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M52	steel	NG	1.038	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.

M49	steel	NG	1.038	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M57	steel	NG	1.038	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M43	steel	WARN	0.926	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M42	steel	WARN	0.926	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M39	steel	WARN	0.922	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M47	steel	WARN	0.922	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M38	steel	WARN	0.922	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M46	steel	WARN	0.922	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M44	steel	WARN	0.920	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.
M41	steel	WARN	0.920	8	Review reserve margin for Deflection.; Review serviceability deflection limit and member stiffness.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	145	40	1.045
Connections	145	50	5.206
Foundations	12	50	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.